90. ZONE 2 TRAINING IS NOT INTRINSICALLY BETTER THAN ZONE 3, 4 AND 5 TO PROMOTE AEROBIC ADAPTATIONS!

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In the last years, there has been a lot of talk about **'zone 2' training, which corresponds to a training intensity below the first lactate/ventilatory threshold.** In fact, to target a specific physiological load, training zones should be divided by specific physiological thresholds like ventilatory thresholds or critical power, given that train and recovery differ if trespassing or not these benchmarks. (1) So, using a classic 5 or 7 zones model, the benchmarks should be placed as following:

- Zone 2-Zone 3 benchmark: around the first ventilatory/lactate threshold;
- Zone 4-Zone 5 benchmark: around the second ventilatory/lactate threshold or critical power (or approximately FTP)

Zone 2 intensity is usually defined as 'easy' to sustain. This makes sense, as no significative muscular metabolic or ionic changes (linked to H+, K+, lactate and phosphocreatine) are observed when riding within this zone. (2). At this intensity the power is almost all produced by the aerobic metabolic pathways.

It has been repeatedly observed with scientific published data that world-class endurance athletes spent 80-90% of the total training time is spent below the first threshold (read zone 1/zone 2). For example:

- in two studies I did during my PhD, we reported that during all the preparation that lead to a top 5 in the final general classification of Giro d'Italia and Tour de France, five world-class road cyclists spent 80-90% of exercise time below 80-85% of FTP, which means, approximately, below the first threshold. (3,4)
- 12 male Norwegian who have been responsible in the past few years for the training of numerous world-class, mostly Norwegian endurance athletes (runners, skiers, rowers, swimmers, cyclists, triathletes) who, in total, have won nearly 400 Olympic-, World-, and European-Championship medals" in , reported they prescribed 80-90% of training time below the first threshold. (5)

It has been speculated that the large amount of low intensity training performed by world-class athletes is due to zone 2 'superpower' for inducing aerobic adaptations, which raise thresholds and VO2max, and so endurance performance.

But, when matching the session energy expenditure, how does Zone 2 training adaptations compare to the higher intensity zones?

When matching the total energy expenditure, does zone 2 stimulate more aerobic adaptations than zone 3, 4, 5, 6 and 7?

A study published by Inglis and Colleagues (University of Calgary, Canada) on *Medicine & Science in Sports & Exercise* in 2024 tried to answer this question. (6)

Answering this question can help to understand if Zone 2 is really ‘intrinsically’ better than the higher intensity zones to induce aerobic adaptations or there may be other reasons behind the large volume at low intensity performed by successful world-class endurance athletes.

WHAT DID THEY DO?

- 84 participants (42 females + 42 males, age ~26, VO2max ~42) completed the study. As you can understand looking at their age and VO2max, they were sedentary/recreationally active and not trained cyclists.
- They were divided into 6 different groups which for 6 weeks trained in the following 6 different ways:
  1. **Zone 2 Group**: 3 times per week 50min at 90% of the first threshold (~65% FTP)
  2. **Zone 3 Group**: 3 times per week ~41 min at 110% of the first threshold (~80% FTP)
  3. **Zone 4 Group**: 3 times per week ~30min at 100% of critical power (~ at FTP)
  4. **Zone 5 Group**: 3 times per week 5/6 x 4min ≥120% of critical power/FTP with 3min recovery in between.
  5. **Zone 6 Group**: 3 times per week 3-6 x30sec all-out sprints with 4.5 min recovery in between preceded by 10min warm-up.
  6. **Control Group:** they did not perform any training

The total work (read energy expenditure) was the same between all the groups (it was matched on purpose) except the Zone 6 group and, of course, the Control Group.

WHAT DID THEY FIND?

VO2max Changes:

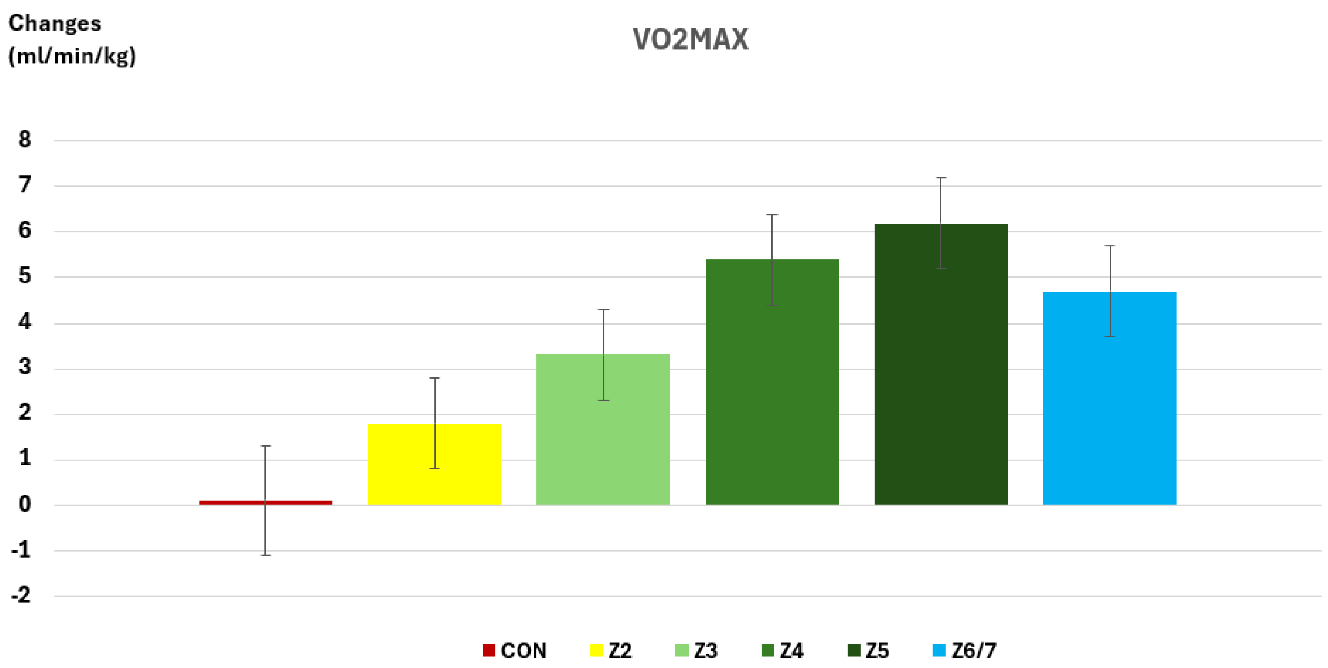
- Control Group: +0.1 ± 1.2 ml/min/kg
- Zone 2 Group: +1.8 ± 2.7 ml/min/kg
- Zone 3 Group: +3.3 ± 2.4ml/min/kg
- Zone 4 Group: +5.4 ± 2.3ml/min/kg
- Zone 5 Group: +6.2 ± 2.8ml/min/kg
- Zone 6 Group: +4.7 ± 2.3ml/min/kg

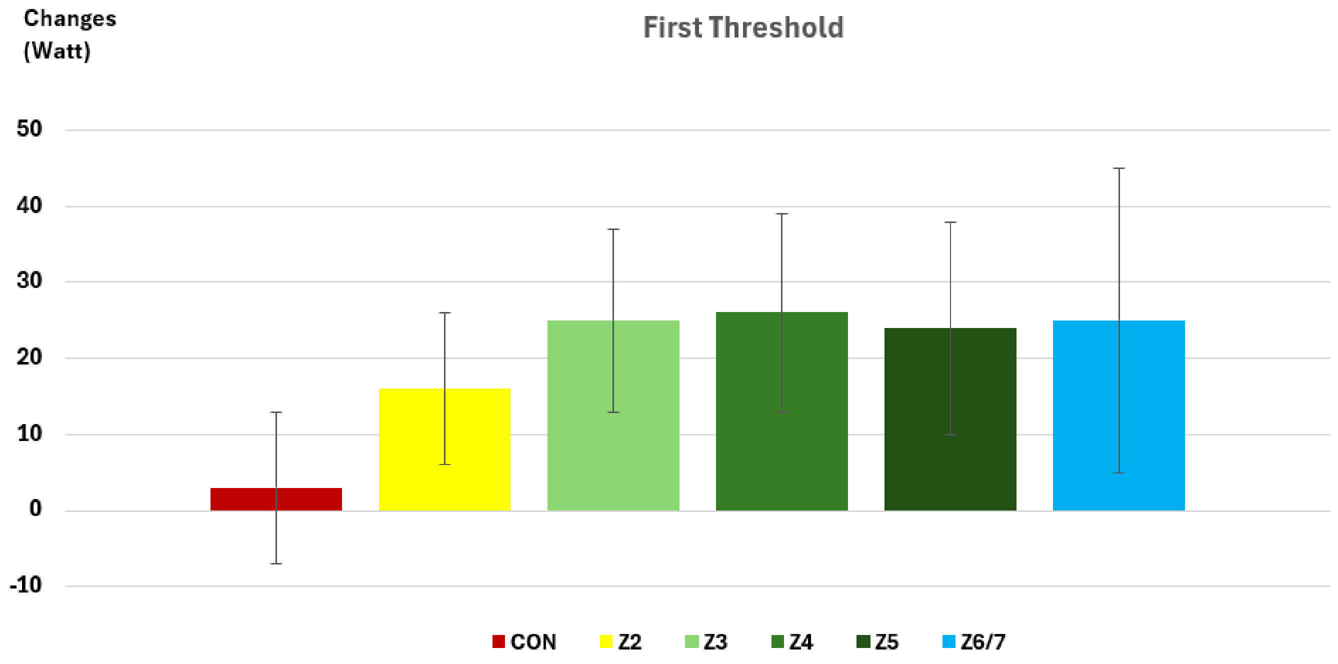
First Threshold Changes:

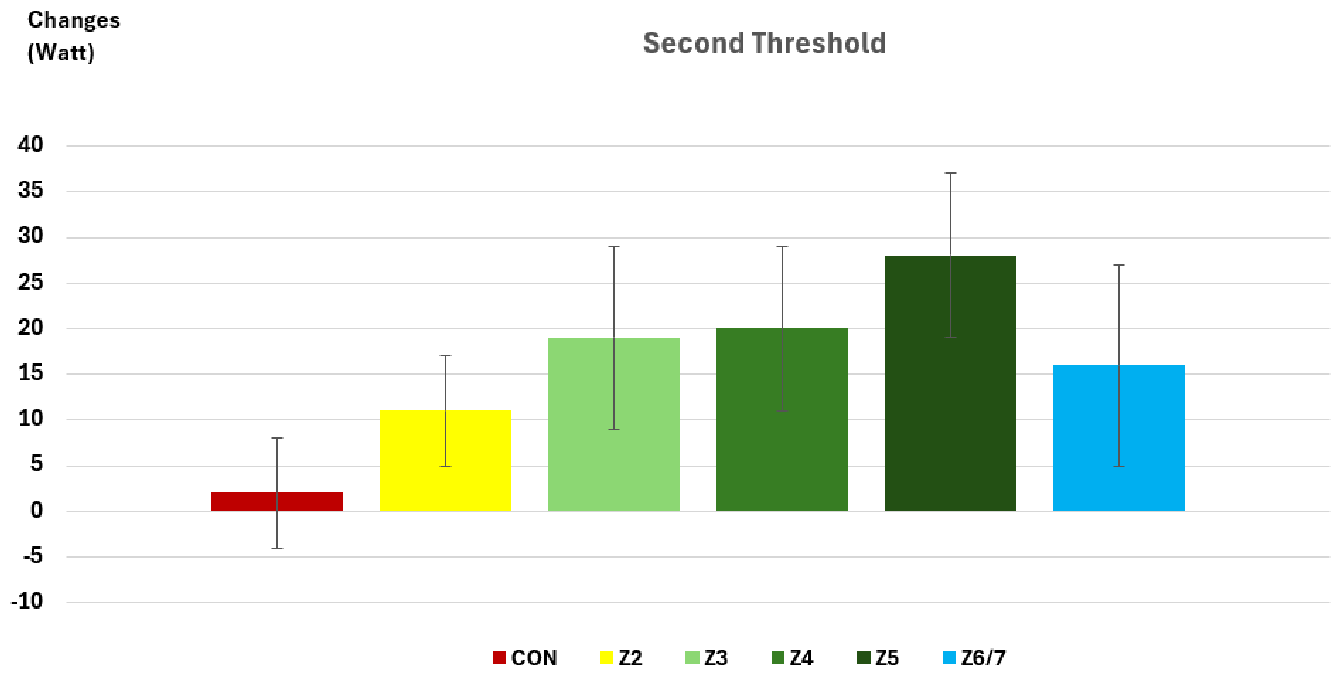
- Control Group: +3 ± 10 Watt
- Zone 2 Group: +16 ± 10 Watt
- Zone 3 Group: +25 ± 12 Watt
- Zone 4 Group: +26 ± 13 Watt
- Zone 5 Group: +24 ± 14 Watt
- Zone 6 Group: +25 ± 20 Watt

Second Threshold Changes:

- Control Group: +2 ± 6 Watt
- Zone 2 Group: +11 ± 6 Watt
- Zone 3 Group: +19 ± 10Watt
- Zone 4 Group: +20 ± 9Watt
- Zone 5 Group: +28 ± 9Watt
- Zone 6 Group: +16 ± 11Watt







Data from Inglis et al. 2024 (6)

PRACTICAL TAKEAWAYS:

- **When matching total work (read energy expenditure), zone 2 induces less aerobic adaptations gains (increase in VO2max and thresholds) when compared to higher exercise intensities (zone 3, 4, 5, 6).**
- **The changes in VO_{2max} and second threshold are positively related to exercise intensity, except for zone 6. In other words, from zone 2 to zone 5, higher the exercise intensity higher the increase in VO_{2max} and second threshold. So, zone 5 seems to be the intensity which is intrinsically better to stimulate aerobic gains.**

- **When evaluating changes in the first threshold, this positive relationship seems to not apply, with zone 3, zone 4 and zone 5 which seems to induce similar gains in power at first threshold.**

GABRIELE'S COMMENT

I think this data can be very helpful to clarify all the talk around zone 2.

Zone 2 is not a magic bullet and it is not intrinsically better than higher intensity zones to induce aerobic adaptations (read improving thresholds and VO2max)!

So, for time crunched cyclists it does not make sense to copy and past the time spent in intensity zones used by successful world-class endurance athletes and spending 80-90% of training time below the first threshold. Based on the above data, my suggestion for these cyclists is doing the 2-3 intensive sessions per week (zone 3, 4, 5, 6) that also pro endurance athletes do and spending less % time in zone 2 compared to world-class athletes.

However, I think this data should not be confounded as 'zone 2 training is not effective at all and so pro athletes are wasting their time'. The above data suggest that higher intensity zones induce more positive aerobic adaptations **when the total energy expenditure is matched**. However, much more energy expenditure can be completed at lower than higher intensity, as higher the intensity higher the stress strain and fatigue which is imposed on the body. So, if you have unlimited time to train, zone 2 is a complementary very effective less stressful way to integrate the limited amount of medium and high intensity work that can be completed within each week/microcycle. So, also for time-crunched cyclists it can be effective to find some ways to accumulate more time in zone 2: bike to work can be a good option if logistically feasible.

Thanks for Reading! Share with your Team!

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